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Volatility forecasting for low-volatility investing

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ABSTRACT

Low-volatility investing often involves sorting and selecting stocks based on retrospective risk measures, for example, the historical standard deviation of returns. In contrast, we employ volatility forecasts from various volatility models to sort, select, and estimate portfolio weights on the 500 largest US stocks. We find that exploiting a large set of time-series models delivers large, significant economic gains compared to traditional benchmarks. After accounting for transaction costs, a low-volatility portfolio based on volatility forecasts from a panel heterogeneous autoregression model and a portfolio based on forecast combinations perform best and can be easily implemented in real time.

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1. Introduction

Low-risk investment strategies are prominent in the financial industry and well motivated by the low-volatility anomaly—the consistent empirical evidence that points towards an inverse relationship between (idiosyncratic) volatility and (risk-adjusted) returns (see Ang et al., 2006; Blitz et al., 2019, among others). Low-volatility portfolios are straightforward investment strategies requiring only estimates of each stock's volatility. Two prominent examples are the S&P 500 Low Volatility Index, which is based on the 100 least volatile stocks in the S&P 500, and the S&P 500 Low Volatility Top 80% Index, which is based on the 400 least volatile stocks in the S&P 500.[1] These indices assign a weight to each stock inversely proportional to the respective stock's volatility, measured by the standard deviation of daily returns over the preceding year. The methodology of the S&P 500 Low Volatility Top 80% Index is motivated by the empirical observation that the relation between volatility and (risk-adjusted) returns is flat for low- and medium-volatility stocks, but negative for the stocks with the highest volatility. Hence, it pays off to avoid investing in high-volatility stocks (Blitz et al., 2019; Gu et al., 2020).

Alternative low-risk strategies are minimum variance portfolios (MVPs). In contrast to low-volatility portfolios, MVPs require estimates of variances and covariances and hence are more challenging to implement, specifically in high dimensions. Kirby and Ostdiek (2012) propose a volatility-timing strategy that only requires the knowledge of variances but not of covariances. Under the constraint that only the 80% stocks with the lowest volatility are considered and when using volatilities instead of variances for computing portfolio weights, the low-volatility portfolios considered in this paper are a special case of the volatility-timing strategy of Kirby and Ostdiek (2012).

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[1] https://www.spglobal.com/spdji/en/documents/methodologies/methodology-sp-low-volatility-indices.pdf

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From an ex ante perspective, it remains an open question whether backward-looking standard deviations of past returns or forward-looking volatility forecasts, e.g., based on high-frequency data, should be used to select stocks and determine the weights of low-volatility portfolios. Our paper focuses on improving low-volatility investing by comprehensively comparing the performance of low-volatility portfolios based on backward-looking risk measures with the performance of portfolios based on model-based forward-looking risk measures. For this, we employ a wide array of volatility models and forecast combinations. Our aim is to provide methodological improvements in implementing low-volatility portfolios with monthly rebalancing. By considering an *infeasible post hoc portfolio* as the benchmark, we take a new perspective on low-volatility allocations. The post hoc portfolio is defined as the portfolio that an investor would choose with perfect foresight of each stock's monthly volatility.

Because the post hoc portfolio is based on portfolio weights that are computed as the inverse of each stock's ex post realized monthly volatility, we can treat the problem of ex ante determining the optimal weights as a prediction problem. In the literature on low-volatility investing, portfolio weights are typically determined based on the backward-looking standard deviation of monthly or daily stock returns over a specific period such as the previous year, the previous six months, or the previous month (see Bali et al., 2016, for an overview of commonly used metrics). At the same time, the financial econometrics literature (see e.g. Ghysels et al., 2019) has developed a wide range of volatility models for anticipating future volatility. If we had access to a volatility model that would perfectly forecast the ex post monthly volatility of each stock, we could implement the post hoc portfolio ex ante. Obviously, due to unavoidable forecast errors, this approach is not feasible. Nevertheless, the better a volatility model's forecast performance, the closer we should get to the post hoc portfolio. This idea links the construction of low-volatility portfolios to the literature on volatility forecasting.

We employ models from different model classes to evaluate which models are best suited for forecasting the one-month-ahead volatility of individual stocks. First, we use simple RiskMetrics models and various generalized autoregressive conditional heteroskedasticity (GARCH)-type models (see e.g. Glosten et al., 1993). Those models treat the conditional variance as a latent process, and daily (or monthly) returns are used for modeling, estimating, and forecasting volatilities. Second, we use heterogeneous autoregression (HAR) models (Corsi, 2009) and mixed-frequency data sampling (MIDAS)-type models (Ghysels et al., 2004). Here, the realized variances are modeled directly as a function of past realized variances. Third, we consider forecast combinations following an approach suggested by Caldeira et al. (2017). The combinations optimally combine model-specific forecasts based on each model's past performance. Importantly, our forecast combination approach is implementable in real time.

In the first step, we follow the financial econometrics literature (see e.g. Ghysels et al., 2019) and evaluate the stock-specific out-of-sample forecast performance of each model. While there is already extensive literature on volatility forecast evaluation, the focus is typically on a relatively small number of assets. Similar to our approach, Ghysels et al. (2019) use GARCH, HAR, and MIDAS models to predict the volatility of 30 assets from different asset classes. Patton and Sheppard (2015) examine a cross-section of 105 assets but cover a considerably smaller spectrum of time-series models, and do not consider forecast combinations. For related studies, see also (Bollerslev, Hood et al., 2018; Bollerslev et al., 2016) and Christensen et al. (2023). Instead, we consider a large cross-section of 1699 US stocks and a forecast horizon of one month for the out-of-sample period from January 2005 to December 2021. Similar to Ghysels et al. (2019), we find that realized variance-based models in combination with direct rather than iterated volatility forecasting outperform all other individual models. However, there is considerable time variation in relative forecast performance. While the Log-HAR performs very well during low volatility periods, models that impose more structure excel during periods of turmoil. During the financial crisis, a HAR-type model that imposes a panel structure (i.e. the same HAR parameters across all stocks) and includes quarterly and semi-annual realized variances (in addition to the usual daily, weekly, and monthly realized variances) performs best for more than 50% of stocks. We refer to this model as Panel HAR-LR. Finally, with the onset of the Covid-19 pandemic, a Factor GARCH-MIDAS model that employs the VIX as a predictor variable is the best-performing model for most stocks. Despite this time variation in relative model performance, individual HAR-type models outperform backward-looking volatility measures over the entire out-of-sample period. For example, for 97% of stocks, the Log-HAR outperforms the backward-looking benchmark measure of volatility based on the standard deviation of daily returns during the previous year (henceforth, the benchmark volatility measure).

We find that forecast combinations further improve forecast performance when assigning equal weights to all forecasts or weights inversely proportional to past forecast losses. This result is in line with the literature on forecast combinations (see e.g. Claeskens et al., 2016; Stock & Watson, 2004).

We then assess which volatility forecasts facilitate constructing low-volatility portfolios as close to the post hoc portfolio as possible. When measuring closeness by the absolute distance between the weights of the post hoc portfolio and a forecast-based portfolio, the Log-HAR achieves the smallest distance. Overall, there is strong evidence that the weights of forecast-based portfolios are closer to the weights of the post hoc portfolio than those of the portfolio based on the backward-looking benchmark volatility measure (henceforth, the benchmark portfolio). When evaluating the portfolio turnover and respective trading costs, the Panel HAR-LR generates the lowest turnover among all forecast-based portfolios. Certain portfolios based on forecast combinations (e.g. the combination that uses equal weights to combine forecasts) lead to further reductions in transaction costs but at the price of a larger distance to the post hoc portfolio.

Finally, we evaluate the economic performance of the forecast-based low-volatility portfolios. We first ignore transaction costs. As expected, the portfolios based on the Log-HAR and the Panel HAR-LR are among the forecast-based portfolios with the lowest tracking error (with respect to the post hoc portfolio). Importantly, a portfolio based on a forecast combination that places all weight on the best-performing model achieves the lowest tracking error. We refer to this as the best-forecast-based (henceforth, BF-based) portfolio. Compared to the volatility of the benchmark portfolio, forecast-based portfolios lead to a significant reduction in portfolio volatility. As a result, the Sharpe ratios of the forecast-based portfolios are significantly higher than the Sharpe ratio of the benchmark portfolio. The BF-based portfolio achieves the highest Sharpe ratio. When accounting for transaction costs, fewer forecast-based portfolios significantly outperform the benchmark. This is because forecast-based portfolios often generate considerably higher turnover than the benchmark. However, the portfolio based on the Panel HAR-LR and the BF-based portfolio still significantly outperform the benchmark regarding risk-adjusted returns. To put the gains that can be achieved into numbers: An investor who switches from the benchmark portfolio to either the Panel HAR-LR-based portfolio or the BF-based portfolio, while maintaining the same level of risk as for the benchmark portfolio, can achieve an increase in average yearly returns of 0.62 percentage points or 0.51 percentage points, respectively, after accounting for transaction costs.

These findings are confirmed when directly comparing the forecast-based portfolios with the post hoc portfolio. When computing the fee that would make a risk-averse investor with quadratic utility function indifferent between the (superior) post hoc portfolio and the forecast-based portfolios, this fee is the lowest for the Panel HAR-LR-based portfolio or the BF-based portfolio. In summary, our results show that volatility forecasting within a large cross-section of stocks can bring important economic gains when used to construct low-volatility portfolios.

The remainder of the paper is organized as follows. Section 2 reviews the previous literature, and Section 3 presents empirical evidence for the efficacy of the inverse volatility allocation. In Section 4.1 the data are presented, and Section 4.2 introduces the volatility models. In Section 5, we show that models with better forecast performance allow us to implement low-volatility portfolios closer to the post hoc portfolio. Finally, Section 6 concludes.

2. Related literature

The anomaly that realized returns appear inversely related to volatility is well documented in the financial literature. Haugen and Heins (1975) make use of total volatility, while Ang et al. (2006) and Ang et al. (2009) study allocations based on idiosyncratic volatility. Here, total volatility refers to the standard deviation of stock-specific returns (e.g. over the previous 252 days), while idiosyncratic volatility refers to the standard deviation of the residuals from a regression of stock-specific returns on the market factor (and potentially other factors such as size and value) (see Bali et al., 2016, Chapter 15). Through the use of machine-learning techniques, Gu et al. (2020) demonstrate that high total volatility predicts low returns. Likewise, there is literature that studies risk defined as beta (i.e. the slope coefficient in a regression of stock-specific returns on the market return) which shows that Sharpe ratios are declining in beta (Frazzini & Pedersen, 2014; Haugen & Heins, 1975). All of these findings are related, as stocks with high market exposures are typically high-volatility stocks, and total volatility is highly correlated with idiosyncratic volatility (Baker et al., 2011; Bali et al., 2016; Blitz et al., 2019).

Several explanations have been proposed for the low-volatility anomaly. Examples are that investors face leverage constraints (e.g. Frazzini & Pedersen, 2014), regulatory constraints, or constraints on short selling (see, for example, Blitz et al., 2014, for an overview). Behavioral explanations include representativeness, overconfidence, or preferences for lottery-like stocks (Baker et al., 2011; Bali et al., 2011; Barberis & Huang, 2008). Asness et al. (2020) find evidence supporting both the leverage and the lottery hypotheses. Schneider et al. (2020) argue that the low-volatility anomaly can be explained by co-skewness risk.

We take a forward-looking perspective on low-volatility portfolios by utilizing time-series models for predicting the volatilities used for computing portfolio weights. This links our paper to the literature on the forecast performance of volatility models (see, for example, Ghysels et al., 2019) and the benefits of using these models for portfolio construction (Bollerslev, Hood et al., 2018; Fleming et al., 2001, 2003). We contribute to this literature by comprehensively studying a large spectrum of volatility models based on daily as well as intraday return data. The potential benefits of using intraday data were investigated by Boudt et al. (2015) but without using volatility models and for a shorter sample of six years.

A significant concern in financial time-series forecasting is the possibility of structural breaks and parameter instability (Conrad & Kleen, 2020). To accommodate these concerns, we consider a rich set of forecast combinations to possibly enhance forecasting accuracy (Rossi, 2021; Timmermann, 2006). Following Bates and Granger (1969), a large spectrum of econometric tools has been developed for the purpose of obtaining optimal forecast combinations (see, for example, Clemen & Winkler, 1986; Palm & Zellner, 1992). The seminal paper by Granger and Ramanathan (1984) estimates the optimal weights in a regression framework. This framework has since been extended into various directions (Diebold & Shin, 2019; Stock & Watson, 2004; Terui & Van Dijk, 2002). Recent surveys of this topic include Timmermann (2006) and Wang et al. (2023). In combining volatility forecasts, we follow the approach suggested by Caldeira et al. (2017), where each forecast's weight is based on past forecast performance. Their flexible aggregation scheme nests the consensus forecast (i.e. each forecast receives the same weight) as well as the case that all weight is given to the best-performing model.

While our focus is on low-volatility portfolios, there are other low-risk investment strategies. For example,

Moreira and Muir (2017) consider volatility-managed portfolios. Those portfolios exploit the persistence of volatility over time: Taking less risk in high-volatility periods can produce significant economic gains. Based on the same relationship, Harvey et al. (1997) argue in favor of the benefits of strategies with volatility targeting. By contributing a comprehensive real-time evaluation and considering volatility-managed portfolios from a multi-factor perspective, Cederburg et al. (2020) and DeMiguel et al. (2021) provide further insights into the workings of volatility-managed portfolios. Finally, the low-volatility portfolios we consider in this paper ignore correlations between stocks. In principle, risk may be further reduced by accounting for correlations and focusing on global minimum-variance portfolios (Bollerslev, Patton et al., 2018). However, empirically, the low-volatility portfolios have several favorable properties, particularly in high dimensions. First, the number of quantities to be estimated is significantly reduced. Second, computing portfolio weights does not require any matrix inversion. Third, short selling is inherently excluded by design.

3. Post hoc portfolio and volatility forecasting

In this section, we take a new perspective on low-volatility allocations by creating and evaluating the performance of an ex ante infeasible post hoc portfolio. As discussed in more detail in Section 4.1, we use daily stock price data from the Center of Research in Security Prices (CRSP) and combine it with intraday data from the New York Stock Exchange TAQ database. Each month, we use the 500 stocks that are the largest as measured by market capitalization. Because this cross-section is time varying between 2005:M1 and 2021:M12, our analysis includes 1699 stocks in total.

First, we provide evidence that the low-volatility anomaly is much stronger from an ex post than an ex ante perspective. We denote the monthly realized variance of stock i, $i = 1, \ldots, N$, in month m by $RV_{i,m}$. The monthly realized variance equals the average of daily realized variances based on intraday data scaled to a monthly level (for details, see Section 4.1). The previous literature on low-volatility portfolios has focused on volatility measures based on daily return data. Following this approach, we measure the variance of stock i at the end of month m by the (appropriately scaled) sum of squared daily returns over the last 12 months (i.e. 252 trading days), which we denote by $\text{12m-RV}^d_{i,m}$.[2] As mentioned in the introduction, we refer to $\text{12m-RV}^d_{i,m}$ as the benchmark volatility measure.

From an ex ante perspective, the low-volatility anomaly can be illustrated by sorting all stocks in month m according to the ascending ordering of $\text{12m-RV}^d_{i,m}$ and forming quintile portfolios for month $m + 1$. We refer to this approach as *trailing volatility sort*. Table 1 shows the Sharpe ratios of the quintile portfolios in month $m + 1$ based on the month m volatility sorting. Sharpe ratios are declining in volatility quintiles: The Sharpe ratios of the quintile portfolios of the 20% stocks with the lowest/highest volatility are the highest/lowest. This confirms a common finding in the literature on the low-volatility effect: Low-volatility stocks have high risk-adjusted returns (see, for example, Boudt et al., 2015). As shown by Moreira and Muir (2017) and Harvey et al. (2018), this effect also holds at the market level. Next, we take an ex post perspective. We think of a hypothetical investor who constructs (infeasible) quintile portfolios formed at the end of month m based on the realized variance of month $m + 1$. We refer to this infeasible approach as *post hoc volatility sort*. As Table 1 shows, the low-volatility effect is much more pronounced from an ex post perspective. Sharpe ratios are now declining from the first to the fourth quintile portfolio and then drop dramatically for the fifth quintile portfolio. Note that Sharpe ratios are now considerably higher/lower for the lowest/highest quintile portfolio than for the trailing volatility sort. A strong negative contemporaneous relation between realized volatility and stock returns drives this result. From an ex post perspective, avoiding high-volatility stocks is most important.

Table 1
Sharpe ratios of equal-weighted quintile portfolios.

Quintile	1	2	3	4	5
Trailing volatility sort	0.89	0.81	0.70	0.51	0.48
Post hoc volatility sort	1.28	0.98	0.72	0.60	0.18

Notes: We report Sharpe ratios based on monthly excess returns of volatility-sorted portfolios. In each month, we sort stocks either based on a trailing volatility proxy (volatility of 12 months of daily return data) or we sort stocks based on the ex post measure of monthly volatility calculated using intraday data. The investment universe is the set of the 500 largest U.S. common stocks by market capitalization in the previous month. The investment period is 2005:M1 to 2021:M12.

Motivated by these observations, we follow the methodology of the S&P 500 Low Volatility Top 80% Index for the remainder of this paper; that is, we exclude the top 20% volatility stocks from our portfolio allocation, and attach weights inversely proportional to the volatilities of the remaining assets (see Section 5.2 for details). We refer to the feasible portfolio that is based on the 80% stocks with the lowest volatility as measured ex ante by the benchmark volatility measure by 12m-RV^d portfolio (or benchmark portfolio). Because it employs the same sorting and weighting, the 12m-RV^d portfolio has a correlation of 98.7% with the S&P 500 Low Volatility Top 80% Index. The small remaining difference is due to the fact that our investment universe is slightly different (the largest 500 common stocks vs. the S&P 500 constituents) and that we have a different rebalancing frequency (monthly vs. quarterly). For comparison, we construct the hypothetical portfolio that is based on the 80% stocks with the lowest $RV_{i,m+1}$ and choose portfolio weights accordingly. In the introduction, we labeled this infeasible portfolio the *post hoc portfolio*.

Fig. 1 shows the cumulative return of the post hoc portfolio (blue), the 12m-RV^d portfolio (purple), and the S&P 500 Low Volatility Top 80% Index (Bloomberg ticker:

[2] Let $r_{i,t}$ denote the daily returns of stock i. Further, let r_{i,T_m} be the return on the last day of month m. Then, we compute $\text{12m-RV}^d_{i,m}$ as $22/252 \sum_{s=0}^{251} r^2_{i,T_m-s}$. There are alternatives to this construction. For example, volatility might be measured using returns over the previous months (e.g. Ang et al., 2006, 2009).

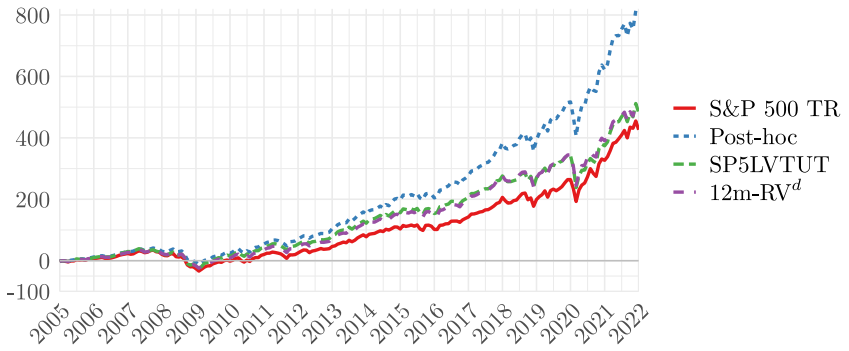


Fig. 1. Cumulative return of volatility-weighted portfolios against market.
Notes: Cumulative returns of three volatility-weighted portfolios along the S&P 500 value-weighted total return from 2005:M1 to 2021:M12. The construction of the low-volatility portfolios mimics the methodology in the S&P 500 Low Volatility Top 80% Index (Bloomberg ticker SP5LVTUT). For more information about the data underlying the low-volatility portfolios, see Section 4.1.

Table 2
Performance statistics for low-volatility portfolios.

	Ret	Std	SR
Post hoc	12.74	12.70	1.00
12m-RV^d	10.52	13.86	0.76
S&P 500 LV Top 80%	10.60	14.40	0.74
S&P 500 TR	10.05	14.63	0.69

Notes: We report arithmetic means of discrete excess returns (Ret), their standard deviation (Std), and the corresponding Sharpe ratio (SR). All measures are annualized. The evaluation period is 2005:M1–2021:M12.

SP5LVTUT, green). The figure also depicts the S&P 500 Index total return (S&P 500 TR, red). Over the whole sample period, the post hoc portfolio outperforms the 12m-RV^d portfolio and the S&P 500 TR. Table 2 presents summary statistics for the four portfolios. The post hoc portfolio has a higher return, a lower volatility and, hence, a considerably higher Sharpe ratio than the 12m-RV^d portfolio. As expected, the performance of the 12m-RV^d portfolio and the S&P 500 Low Volatility Top 80% Index is very similar. In comparison to the S&P 500 TR, the 12m-RV^d portfolio mainly benefits from a reduction in volatility.

In what follows, we aim at constructing low-volatility portfolios that come closer to the post hoc portfolio than the 12m-RV^d portfolio. We treat the task of replicating the post hoc portfolio as a forecasting problem. The volatility forecast of stock i in month $m+1$ based on information up to the end of month m is denoted by $\widehat{RV}_{i,m+1|m}$. As discussed in the introduction, a 'perfect' volatility model would correctly predict the volatility of all N stocks in month $m+1$, i.e. $\widehat{RV}_{i,m+1|m} = RV_{i,m+1}$. Based on those 'perfect forecasts', it would be possible to construct the post hoc portfolio ex ante. In practice, forecast errors are unavoidable. Nevertheless, the better the forecast performance of a volatility model, i.e., the lower the forecast losses, the closer the forecast-based low-volatility portfolio should mimic the post hoc portfolio. We address two critical questions. First, do volatility models lead to portfolio allocations closer to the post hoc portfolio than portfolios based on backward-looking measures (such as the 12m-RV^d portfolio)? On the one hand, state-of-the-art volatility models are known to provide accurate volatility forecasts (Ghysels et al., 2019). On the other hand, when applied to a large cross-section of stocks, the estimation error inherent in volatility models, may lead to a portfolio allocation that is inferior to an allocation based on a simple backward-looking measure. The intuition is similar to that of DeMiguel et al. (2009), who show that estimation error is often large enough to offset the gains obtained through econometric methods in a portfolio optimization context: The simple $1/N$ portfolio often outperforms 'optimal' portfolios based on estimated weights. Second, extending the work of Boudt et al. (2015), we investigate whether it pays off to employ high-frequency based measures of volatility instead of measures based on daily return data.

Ultimately, we address the forecasting problem in three steps. We first estimate various volatility models for each stock and evaluate the forecast performance of each model (see Section 5.1). Fig. 2 highlights the persistence of the realized volatilities in the cross-section of assets included in our analysis. This persistence generates the predictability that state-of-the-art volatility models exploit. This first step allows us to answer the question of which volatility models provide the best forecasts of monthly stock volatility in a large cross-section of stocks.

Secondly, we evaluate whether the forecasts from the volatility models translate into more accurate volatility weights than those from the benchmark measures (see Section 5.2). Recall that we assign a zero weight to the 20% stocks with the highest volatility forecasts and weights inversely proportional to the volatility forecasts to the 80% remaining stocks. Accuracy is measured by the absolute deviations of the forecast-based weights from the weights of the post hoc allocation.

Lastly, we evaluate the out-of-sample performance of the portfolio allocations (see Section 5.3). We assess how attractive the portfolios are to investors in terms of both gross performance and net of transaction costs. To make this evaluation as realistic as possible, we estimate the effective costs of trading each asset.

Finally, there is an important caveat. While a perfect volatility model would allow us to implement the post hoc portfolio ex ante, this is also true for certain misspecified models. For example, a model with predictions

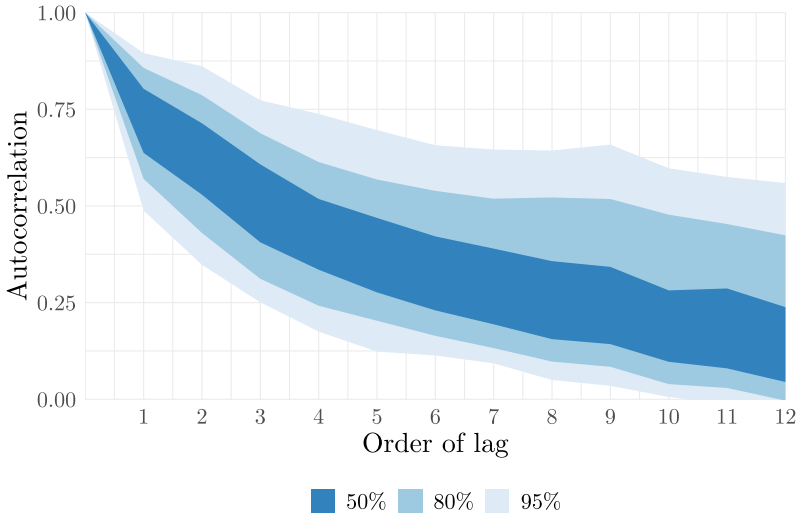


Fig. 2. Autocorrelation of monthly realized variances.
Notes: The fan chart depicts the empirical autocorrelation functions (ACFs) of monthly realized variances across 582 stocks that (1) were among the 500 largest at some point from 2001:M1 to 2021:M12 and (2) were continuously traded during this period. The ACF is estimated using the instrumental variables regression proposed by Hansen and Lunde (2014) with one to six lagged monthly RVs as instruments.

$\widehat{RV}_{i,m+1|m} = \phi RV_{i,m+1}$, for all i, with $0 < \phi \neq 1$ can make large forecast errors but nevertheless allows us to implement the post hoc portfolio correctly.

4. Data and models

4.1. Data

Monthly portfolio returns are calculated from monthly total returns from the Center of Research in Security Prices (CRSP). For the investment universe, we include stocks among the largest 500 as measured by market capitalization in at least one month, between January 2005 and December 2021. This leaves us with a time-varying cross-section of 1669 stocks in total. To start the portfolio analysis in 2005, we extend the data for the stocks in our investment universe back to 1997 such that we have eight additional years of observations before the beginning of the out-of-sample period.

We adjust for CRSP delisting returns to have a survivorship bias-free data set (Bali et al., 2016; Shumway, 1997). Similar to Bollerslev et al. (2020) and Bollerslev et al. (2022), we merge daily CRSP data with NYSE TAQ intraday data. Open and close prices per day are taken from the daily CRSP data files. All intraday transaction data are obtained from NYSE TAQ. These trade data are cleaned according to Barndorff-Nielsen et al. (2009) and we include only trades from the exchange that is referenced in the daily CRSP data. The two data sets are merged via the Wharton Research Data Services linking tables.

Some of our models rely on intraday market data. For this, one-minute intraday data for the S&P 500 are downloaded from Tick Data.[3] Daily values for the VIX are obtained from the Cboe website.[4] For the intraday realized variance estimates, we include prices during market hours from 09:30 to 16:00 and calculate five-minute log returns. The first five-minute return of each day is an open-to-close return, and all others are close-to-close ones. We use five-minute returns, since these are common in the literature and because it has been shown to be a fairly robust choice as a tradeoff between using high-frequency data and obstructing micro-structure noise-related estimation errors (Liu et al., 2015). To further strengthen our volatility proxy, we calculate five-minute returns across five grids, each starting at 09:30, 09:31, 09:32, 09:33, or 09:34. The five resulting daily five-minute RV proxies are then averaged. We rescale the intraday-based realized variance to the daily close-to-close period, as discussed by Hansen and Lunde (2006), where we use an increasing window to assess the ratio between squared daily close-to-close returns and intraday RV. On day t and for stock i, we denote this combined measure by $RV_{i,t}$. The monthly realized variance, $RV_{i,m}$, of stock i is defined as the average $RV_{i,t}$ over all days t in month m times 22, such that it is measured on a monthly scale. Alternatively, squared daily (close-to-close) returns are often used as a simple but less accurate measure of volatility. We denote this noisy proxy by $RV^d_{i,t}$.

The monthly risk-free rate $R_{rf,m}$ is obtained from Kenneth R. French's data library.[5] We downloaded real-time housing starts data from ALFRED[6] and term spread data from the website of the New York Federal Reserve.[7]

[3] https://www.tickdata.com

[4] http://www.cboe.com/products/vix-index-volatility/vix-options-and-futures/vix-index/vix-historical-data

[5] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/

[6] https://alfred.stlouisfed.org/series?seid=HOUST

[7] https://www.newyorkfed.org/research/capital_markets/ycfaq.html#/

4.2. Volatility models

We examine a selection of 24 models that are commonly used for modeling and forecasting volatility. These models can be broadly classified into four categories: RiskMetrics, GARCH, HAR, and MIDAS. The RiskMetrics and GARCH models treat volatility as a latent variable, while the HAR and MIDAS specifications directly model the realized variances. Next, we provide a brief introduction to each of the model specifications, with a more comprehensive description available in Appendix A.

We use four variants of the RiskMetrics model, which is an exponentially weighted moving average (EWMA). Two variants employ monthly realized variances based on squared daily returns, while the other two employ weighted averages of squared daily returns. The RiskMetrics models also differ in the amount of historical return data they consider, with some models using either six or 12 months of data. It is worth noting that the RiskMetrics models can be viewed as restricted GARCH models with fixed ARCH/GARCH parameters and a constant equal to zero.

Besides the simple GJR-GARCH of Glosten et al. (1993), we employ a Panel GARCH model that uses variance targeting for each stock and restricts the ARCH/GARCH coefficients to be the same across stocks. We also combine the Factor GARCH model of Engle et al. (1990) with the GARCH-MIDAS of Engle et al. (2013). We use the VIX, housing starts, and the term spread as explanatory variables in the long-term component. Those variables are powerful predictors of longer term aggregate stock market volatility (Conrad & Kleen, 2020; Conrad & Loch, 2015). Correspondingly, these models are denoted as Factor GARCH-VIX, Factor GARCH-ΔHous, and Factor GARCH-TS. We also consider two types of multiplicative error models (MEMs) (Engle & Gallo, 2006). The MEMs differ from other GARCH models because they are estimated using the square root of realized variances instead of daily return data.

We consider the original HAR specification of Corsi (2009) as well as nine extensions. In the original HAR model the realized variance is a linear function of the lagged daily, weekly, and monthly realized variances. Among the extensions are specifications that model the realized variance of stock i as depending on stock i's lagged realized variances but also on a HAR-type forecast of the S&P 500's volatility or the VIX index. Moreover, it has been documented that pooled cross-sectional estimation improves the out-of-sample forecast performance; see, for example, Bollerslev, Hood et al. (2018) and Kleen et al. (2022). We denote these models by Panel HAR(-LR). Additionally, we include two alternative HAR specifications. First, we include a restricted HAR (R-HAR) with parameter restrictions such that we have a coefficient of one in front of the monthly lagged RV and all other coefficients are equal to zero. The R-HAR corresponds to a random walk model for the monthly realized variance. Second, we also include the Log-HAR model, which applies a logarithmic transformation to the daily, weekly, and monthly RVs.

The MIDAS class of volatility models was proposed by Ghysels et al. (2004, 2005, 2006). In this approach, the realized variance is modeled as a weighted average of lagged daily realized variances, where the weights are parsimoniously parameterized via a flexible parametric weighting scheme. The HAR model of Corsi (2009) is nested when imposing certain constraints on the weights.

Table 3 provides an overview of all models. In addition, it describes four backward-looking volatility measures. Besides our benchmark volatility measure (12m-RV^d), we use backward-looking measures based on returns over the previous four years (4y-RV^d), six months (6m-RV^d), and one month (1m-RV^d). Those measures are computed similarly as described in footnote 2 for 12m-RV^d. Starting in 1997, we estimate all volatility models on a rolling window of four years. This allows us to generate four years of out-of-sample forecasts (from January 2001 to December 2004) before constructing portfolio weights from 2005 onward. The only exceptions are the three variants of Factor GARCH-MIDAS models. For those models, the rolling window is longer. We employ housing starts or term spread data beginning in 1987 and the VIX and S&P 500 returns beginning in 1990 to identify the long-term component.

Ghysels et al. (2019) compare iterated versus direct multi-step-ahead forecasting for GARCH, HAR, and MIDAS models. Following their recommendations, we directly forecast the average 22-day realized variance for all HAR-type and MIDAS models but compute iterative volatility forecasts for the GARCH models and MEMs.

5. Forecast-based low-volatility portfolios

In this section, we first evaluate the forecast performance of the various volatility models relative to the benchmark (i.e. 12m-RV^d) and other backward-looking measures. We then investigate whether better forecast performance allows for constructing low-volatility portfolios closer to the post hoc portfolio. Finally, we evaluate the economic performance of the model-based portfolios relative to the benchmark and the post hoc portfolio. Differences in performance are evaluated from the perspective of a risk-averse mean–variance investor.

5.1. Statistical forecast evaluation

In the first step, we statistically evaluate the forecast performance of the different benchmark measures and volatility models, $j = 1, \ldots, J$. Because the true volatility of each stock i in month m is unobservable, we evaluate model j's out-of-sample forecasts, $\widehat{RV}^j_{i,m|m-1}$, against $RV_{i,m}$, an unbiased volatility proxy. Following Patton (2011), we rely on robust loss functions $L(RV_m, \widehat{RV}^j_{m|m-1})$ for the evaluation. This is because the ranking of two competing forecasts is preserved for robust loss functions when an unbiased proxy replaces the true but unobservable volatility. As shown by Patton (2011), the squared error (SE) loss and the QLIKE loss are two robust loss functions. We mainly rely on the QLIKE loss defined as

$$L(RV_{i,m}, \widehat{RV}^j_{i,m|m-1}) = RV_{i,m}/\widehat{RV}^j_{i,m|m-1} - \log(RV_{i,m}/\widehat{RV}^j_{i,m|m-1}) - 1$$

Table 3
Methods included in the empirical evaluation.

Model	Description
12m-RV^d	Historical variance based on 12 months of daily data
4y-RV^d	Historical variance based on 4 years of daily data
6m-RV^d	Historical variance based on 6 months of daily data
1m-RV^d	Historical variance based on 1 month of daily data
RM monthly, 12 months	EWMA (Morgan, 1996) based on 12 months of monthly data
RM monthly, 6 months	EWMA (Morgan, 1996) based on 6 months of monthly data
RM daily, 12 months	EWMA (Morgan, 1996) based on 12 months of daily data
RM daily, 6 months	EWMA (Morgan, 1996) based on 6 months of daily data
GJR-GARCH	Glosten et al. (1993)
Panel GJR-GARCH	Pakel et al. (2011)
Factor GARCH	Engle et al. (1990)
Factor GARCH-VIX	Factor GARCH using VIX as predictor
Factor GARCH-ΔHous	Factor GARCH using housing starts as preditor
Factor GARCH-TS	Factor GARCH using term spread as predictor
MEM	Engle and Gallo (2006)
Panel MEM	Equivalent extension as Panel GJR-GARCH
R-HAR	Random walk for RV
HAR	Corsi (2009)
HAR-LR	Adapted HAR to monthly horizon
HAR-SPX	Includes a variance forecast for the SPX
HAR-SPX-LR	Adapted HAR-SPX to monthly horizon
HAR-VIX	Bekaert and Hoerova (2014)
HAR-VIX-LR	Adapted HAR-VIX model to monthly horizon
Panel HAR	Bollerslev, Hood et al. (2018)
Panel HAR-LR	Adapted Panel HAR model to monthly horizon
Log-HAR	HAR model after applying logarithmic transformation
MIDAS	Ghysels et al. (2004)
Panel MIDAS	Equivalent extension as Panel GJR-GARCH

Notes: A detailed description of the models can be found in Appendix A.

for the forecast evaluation. The SE loss is given by

$$L(RV_{i,m}, \widehat{RV}^j_{i,m|m-1}) = (RV_{i,m} - \widehat{RV}^j_{i,m|m-1})^2.$$

We prefer the QLIKE because it is less sensitive to outliers than the SE and hence more reliable (see Patton, 2011). As a robustness check, we include results for the SE loss in Table B.1 in the appendix. For each model j, we define the average time-series loss over the M out-of-sample observations for stock i during the 2005:M1–2021:M12 period as

$$L^j_i = \frac{1}{M}\sum_{m=1}^{M} L(RV_{i,m}, \widehat{RV}^j_{i,m|m-1}). \quad (1)$$

To align the forecast evaluation with the investment analysis, we only consider forecasts used to form portfolio weights, i.e. forecasts for stocks among the 500 largest stocks. Thus, M is $17 \times 12 = 204$ months at maximum. For stocks that are not always among the 500 largest stocks, M will be less than 204. For brevity of notation, we refrain from introducing a stock-specific M. The losses in Eq. (1) are the stock-specific losses commonly reported in the volatility forecasting literature (see, for example, Ghysels et al., 2019). We denote the loss of the benchmark volatility measure (i.e. 12m-RV^d) by L^B_i. In Panel A of Table 4, we evaluate the forecast performance of the backward-looking measures, the RiskMetrics models and the model-based forecasts using the QLIKE loss. First, we report

$$LR^j = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{L^j_i/L^B_i<1}, \quad (2)$$

that is, the share of stocks for which model j outperforms the benchmark volatility measure. The 'LR^j' column shows that the benchmark 12m-RV^d is outperformed by essentially all RiskMetrics, GARCH-type, HAR, and MIDAS models for most stocks. Interestingly, the R-HAR (restricted HAR) has a LR^j of 93%. In contrast, the 1m-RV^d achieves an out-performance rate of only 63%, illustrating the benefits of using high-frequency instead of daily data for measuring monthly variances. By combining high-frequency data with a dynamic specification, the Panel HAR, Panel HAR-LR, Log-HAR, and MIDAS lead to further improvements. The best model is the Log HAR, which achieves an out-performance rate of 97%.

Next, we focus on the L^j_i/L^B_i ratio. This ratio tells us by how much the forecast loss can be reduced when replacing the benchmark volatility measure with model j's forecast. The 'Mean' and 'Med' columns report the mean and median of the ratio across stocks. The two columns show that the best model-based forecasts can reduce the loss by more than 40% (on average). In the last two columns, we report for how many stocks a specific model j is ranked best in terms of L^j_i (denoted by $Rk^j = 1$) or is among the top four models (denoted by $Rk^j \leq 4$). Based on this metric, the Panel HAR-LR performs best. Again, the R-HAR, Log-HAR, and HAR-VIX perform well. Interestingly, the Factor GARCH-MIDAS models, particularly the Factor GARCH-VIX, are among the best models. Notably, many top-four models impose a panel structure or feature the VIX as an explanatory variable.[8]

Table 4
Evaluation of forecast performance.

Model	LR^j	Mean	Med	$Rk^j = 1$	$Rk^j \leq 4$
Panel A: Individual forecasts					
12m-RV^d	–	–	–	0.00	0.01
4y-RV^d	0.14	1.45	1.32	0.00	0.00
6m-RV^d	0.79	0.96	0.91	0.00	0.01
1m-RV^d	0.63	1.12	0.87	0.00	0.02
RM monthly, 12 months	0.91	0.98	0.97	0.00	0.01
RM monthly, 6 months	0.81	0.95	0.89	0.00	0.01
RM daily, 12 months	0.89	0.83	0.78	0.00	0.05
RM daily, 6 months	0.88	0.85	0.78	0.01	0.05
GJR-GARCH	0.82	0.86	0.79	0.02	0.04
Panel GJR-GARCH	0.69	0.96	0.89	0.00	0.01
Factor GARCH	0.85	0.74	0.67	0.05	0.22
Factor GARCH-VIX	0.90	0.69	0.64	0.17	0.32
Factor GARCH-ΔHous	0.91	0.71	0.66	0.01	0.21
Factor GARCH-TS	0.90	0.72	0.67	0.00	0.13
MEM	0.03	5.92	5.09	0.00	0.01
Panel MEM	0.02	6.02	5.49	0.00	0.00
R-HAR	0.94	0.62	0.58	0.08	0.25
HAR	0.88	0.68	0.61	0.01	0.12
HAR-LR	0.66	1.40	0.80	0.00	0.07
HAR-SPX	0.80	0.74	0.61	0.06	0.24
HAR-SPX-LR	0.66	1.46	0.78	0.01	0.08
HAR-VIX	0.85	0.70	0.57	0.09	0.37
HAR-VIX-LR	0.63	1.56	0.80	0.02	0.11
Panel HAR	0.96	0.58	0.55	0.01	0.28
Panel HAR-LR	0.97	0.55	**0.52**	**0.24**	**0.58**
Log-HAR	**0.97**	**0.54**	0.53	0.17	0.50
MIDAS	0.96	0.60	0.58	0.01	0.16
Panel MIDAS	0.94	0.62	0.58	0.02	0.13
Panel B: Averaged forecasts					
$\eta = 0$	0.99	0.59	0.56	–	–
Panel C: Combined forecasts $\delta = 12$					
$\eta = 1/2$	0.99	0.55	0.52	–	–
$\eta = 1$	**0.99**	**0.52**	**0.50**	–	–
$\eta = \infty$	0.94	0.59	0.52	–	–
Panel D: Combined forecasts $\delta = 48$					
$\eta = 1/2$	0.99	0.56	0.53	–	–
$\eta = 1$	0.99	0.54	0.52	–	–
$\eta = \infty$	0.91	0.64	0.55	–	–

Notes: For the QLIKE loss function, the 'LR^j' column reports the proportion of stocks in which the stock-specific loss L_i^j of model j is lower than that of the 12m-RV^d benchmark forecast. The second (third) column reports the time-series mean (median) of the loss ratio L_i^j/L_i^B. $Rk^j = 1$ and $Rk^j \leq 4$ report the proportion of the respective model being the best or among the four best-performing models as measured by L_i^j. In Panels B–D, we report results for combined forecasts for different lengths of forecast combination windows where δ corresponds to the number of months to determine optimal forecast weights. In Panel A, numbers in bold highlight the lowest average and median loss ratio across models. Similarly, we highlight the highest LR^j, $Rk^j = 1$, and $Rk^j \leq 4$. We jointly apply the same highlighting to Panels B–D, which are based on forecast combinations. We report all figures up to two decimals. All loss statistics are solely based on (1) forecasts that are used to form portfolios and (2) stocks that are included in the portfolios at least 24 times. The evaluation period is from 2005:M1–2021:M12.

We complement the evidence for the entire out-of-sample period in Table 4 with rolling window evidence of the performance of the models over time. For this, we define $\bar{L}^j_{i,m-1}$ as

$$\bar{L}^j_{i,m-1} = \frac{1}{\delta}\sum_{s=1}^{\delta} L^j(RV_{i.m-s}, \widehat{RV}_{i,m-s|m-1-s}). \quad (3)$$

That is, in month $m-1$, $\bar{L}^j_{i,m-1}$ measures the performance of model j for stock i by the average loss over the last δ periods. For computing $\bar{L}^j_{i,m-1}$ from January 2005 onward, we now also employ the forecasts for the January 2001 to December 2004 period. For the models which were the most often among the top four in Table 4, Fig. 3 plots the share of stocks for which those models were among the top four over time based on $\bar{L}^j_{i,m-1}$ and $\delta = 48$. The figure reveals some time variation in relative forecast performance. In particular, the Log-HAR did very well at the

[8] Table B.1 in the appendix shows that all results are robust to replacing the QLIKE with the SE loss.

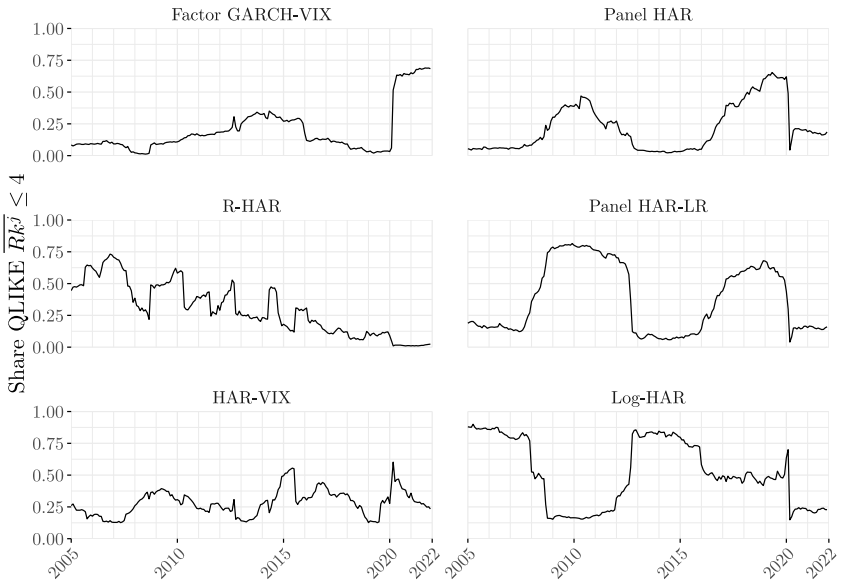


Fig. 3. Top-four ranking across stocks over time.
Notes: We depict the share of stocks for which the respective model is among the top four models according to $L^j_{i,m}$ with $\delta = 48$. For each stock and month, we calculate QLIKE losses based on the trailing 48 months. After this, we rank the models in each month based on these QLIKE losses. The evaluation period is from 2005:M1–2021:M12.

beginning of the out-of-sample period. The Panel HAR-LR performed best during and after the financial crisis, and the Factor GARCH-VIX has outperformed since the beginning of the Covid-19 pandemic.

The findings from Fig. 3 suggest that further improvements in forecast performance can be achieved by considering forecast combinations.[9] We construct combined forecasts for the volatility of stock i, $i = 1, \ldots, N$, in period m based on month $m-1$ information as

$$\widehat{RV}^{cf}_{i,m|m-1} = \sum_{j=1}^{J} \lambda^j_{i,m-1} \widehat{RV}^j_{i,m|m-1}, \tag{4}$$

where the weights $\lambda^j_{i,m-1}$ are positive and sum up to one. Following Caldeira et al. (2017), we specify the weights as

$$\lambda^j_{i,m-1} = \frac{(\bar{L}^j_{i,m-1})^{-\eta}}{\sum_{s=1}^{J} (\bar{L}^s_{i,m-1})^{-\eta}} \tag{5}$$

with $\eta \geq 0$. By setting $\eta = 0$, we assign equal weights to the forecasts. The resulting average forecast is often called the consensus forecast. When $\eta = \infty$, we assign a weight of one to the model with the lowest loss in Eq. (3) and a zero weight to all other models. Weights are inversely proportional to the loss of the respective model when $\eta = 1$. Note that $\eta = 1/2$ in combination with the SE implies that the weights are chosen according to the root mean squared error. In Panel B of Table 4, we set $\eta = 0$. In Panels C and D, we use $\delta = 12$ and $\delta = 48$; that is, we average over the previous 12 or 48 months. Panels B–D show that forecast combinations with $\eta \in \{0, 0.5, 1\}$ lead to further improvements by achieving outperformance ratios of 99%. Fig. 4 is based on the $\delta = 48$ and $\eta = \infty$ combination, which places all weight on the $Rk^j = 1$ model. For the same models as in Fig. 3, the figure shows the fraction of stocks for which the respective models perform best. In addition, we include the benchmark volatility measure and, as a measure of market turbulence, the monthly realized volatility of the S&P 500. There are several interesting insights. First, a single model can be the best-performing model for over 50% of stocks over several consecutive years. Second, periods of market turmoil (as measured by the S&P 500's volatility) coincide with changes in top-performing models. The R-HAR and the Log-HAR did very well during the low-volatility period before the financial crisis. With the beginning of the financial crisis, the Panel HAR-LR became the dominant model. When volatility decreased after the European sovereign debt crisis, the Log-HAR took over again. With the outbreak of the Covid-19 pandemic, the Factor GARCH-VIX performed very well. Finally, there are almost no stocks for which the 12-RV^d benchmark is the best-performing model.

5.2. How close do we get to the post hoc portfolio and at which price?

Next, we compare low-volatility portfolios based on either backward-looking measures or volatility forecasts. Specifically, we evaluate whether portfolios based on model-based volatility forecasts 'come closer' to the post hoc portfolio than the benchmark portfolio. We also investigate whether 'closeness' comes at the cost of (excessive) turnover and trading costs.

[9] Bates and Granger (1969) were the first to promote forecast diversification by combining forecasts from different models. For further discussions, see, for example, Timmermann (2006).

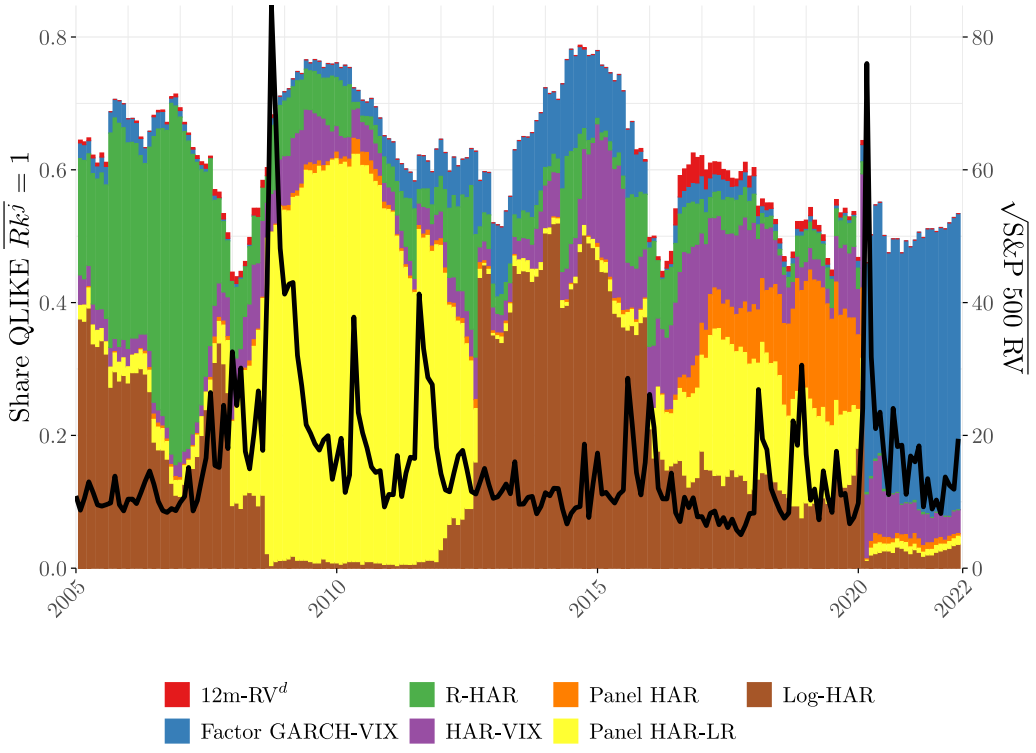


Fig. 4. Top-one ranking across stocks over time.
Notes: The shaded areas in this plot represent the share of the 500 stocks for which the corresponding model ends up being the best-performing model (left axis). For each stock and month, we calculate QLIKE losses based on the trailing 48 months. After this, we rank the models based on these losses. This model ranking is used for the forecast combination with $\eta = \infty$ and $\delta = 48$. The evaluation period is from 2005:M1–2021:M12. The black like represents the monthly realized volatility of the S&P 500 (right axis).

We illustrate the construction of the low-volatility portfolios based on the volatility forecasts of model j. Assume that the volatility forecasts $\widehat{RV}^j_{i,m|m-1}$ for the $i = 1, \ldots, N$ stocks in month m are already in ascending order; that is, $\widehat{RV}^j_{1,m|m-1} \leq \widehat{RV}^j_{2,m|m-1} \leq \cdots \leq \widehat{RV}^j_{N,m|m-1}$. Based on this ordering of the forecasts, the 80% stocks with the lowest volatility are included in the portfolio for month m. Following the construction of the S&P 500 Low Volatility Top 80% Index, stocks receive weights inversely proportional to their forecasted volatility. We denote the individual weight of stock i with respect to the forecasts from model j by

$$w^j_{i,m} \propto \frac{1}{\sqrt{\widehat{RV}^j_{i,m|m-1}}}.$$

The weights are standardized such that the sum of all weights is equal to one. All remaining stocks in the highest volatility quintile receive a weight of zero.

Weight absolute distance (WAD): We measure the distance to the infeasible post hoc portfolio allocation in month m by the absolute deviation of the forecast-based weights and the ex post optimal weights as

$$WAD^j_m = \sum_{i=1}^{N} |w^j_{i,m} - w^{ph}_{i,m}|, \tag{6}$$

where $w^{ph}_{i,m}$ denotes the weights of the post hoc portfolio. Note that WAD^j_m includes all stocks, i.e. those that receive a positive weight (included stocks) and those with zero weight (excluded stocks). Table 5 reports the average distance for each model j, defined as $WAD^j = M^{-1} \sum_{m=1}^{M} WAD^j_m$. The WAD^j of the benchmark 12m-RV^d is 33.44%. That is, the average monthly overlap with the weights of the post hoc portfolio is 77%. The WAD^j of the other backward-looking measures, the RiskMetrics models, as well as the GARCH-type models take similar values. In sharp contrast, the portfolio allocation of the HAR models is much closer to the portfolio allocation of the post hoc portfolio. For the Log-HAR, the WAD^j is 19.75%. That is, the Log-HAR achieves an average overlap of 80%.

Turnover: Recall that $w^j_{i,m}$ is the weight assigned to stock i by model j at the end of month $m-1$. Based on the volatility forecasts for month $m+1$, the new desired weights are $w^j_{i,m+1}$. Before the next rebalancing at the end of period m, due to price movements, the weight of stock i has changed to $w^j_{i,m} \frac{1+R_{i,m}/100}{1+(w^j_m)' R_m/100}$, where $w^j_m = (w^j_{1,m}, \ldots, w^j_{N,m})'$, and R_m is the vector of equity returns; that is, $R_m = (R_{1,m}, \ldots, R_{N,m})'$. Hence, the turnover due to

portfolio rebalancing at the end of month m is given by

$$TO_m^j = \sum_{i=1}^{N} TO_{i,m}^j = \sum_{i=1}^{N} \left| w_{i,m+1}^j - w_{i,m}^j \frac{1 + R_{i,m}/100}{1 + (w_m^j)' R_m/100} \right|. \tag{7}$$

The quantity TO_m^j can be interpreted as the proportion of wealth reallocated at the end of month m. This approach follows the literature on portfolio allocation based on high-frequency-based measures of realized (co-)variation (Bandi et al., 2008; De Pooter et al., 2008; DeMiguel et al., 2009; Hautsch et al., 2015; Nolte & Xu, 2015).

Transaction costs: To assess the impact of transaction costs on portfolio performance, we follow the approach of Novy-Marx and Velikov (2016). Effective costs $c_{i,m}$ per stock and month are estimated through the generalized trading-cost model proposed by Hasbrouck (2009). We calculate the transaction costs of forecast-based portfolio j in month m as $TC_m^j = \sum_{i=1}^{N} c_{i,m} TO_{i,m}^j$.

In Table 5, we report TO^j, which we measure by the time average of the fraction of wealth reallocated at rebalancing, as well as TC^j, which we measure by the time average of TC_m^j. The table shows that the 12m-RV^d benchmark portfolio has low turnover and, consequently, low trading costs. Portfolios based on the forecasts of individual models can have a turnover that is twice as high. Imposing the panel assumption reduces turnover considerably. Similarly, forecast combinations with $\delta \geq 12$ and low η reduce turnover and trading costs. However, it is important to note that the post hoc portfolio also has high turnover (24.93) and trading costs (4.92). The turnover of the forecast-based portfolios with the lowest WAD^j is similar to that of the post hoc portfolio. Nevertheless, there is a tradeoff. When replacing backward-looking volatility measures with volatility forecasts, the distance to the post hoc allocation is considerably reduced, but higher turnover and trading costs will negatively affect portfolio performance.

Table 5
Portfolio characteristics.

Post-hoc	WAD	TO	TC
	–	24.93	4.92
Panel A: Portfolios based on individual forecasts			
12m-RV^d	33.44	9.99	2.00
4y-RV^d	35.96	8.34	1.69
6m-RV^d	33.41	13.07	2.58
1m-RV^d	37.25	42.89	8.23
RM monthly, 12 months	33.27	10.00	2.00
RM monthly, 6 months	33.33	13.09	2.58
RM daily, 12 months	33.84	20.09	3.91
RM daily, 6 months	33.77	20.58	3.98
GJR-GARCH	34.29	20.14	4.16
Panel GJR-GARCH	34.70	16.46	3.48
Factor GARCH	34.56	14.68	3.10
Factor GARCH-VIX	35.71	15.31	3.22
Factor GARCH-ΔHous	35.72	15.45	3.24
Factor GARCH-TS	35.74	15.41	3.24
MEM	31.55	19.46	4.00
Panel MEM	26.62	17.58	3.71
R-HAR	20.31	23.80	4.65
HAR	23.21	16.72	3.52
HAR-LR	24.97	18.65	4.07
HAR-SPX	24.55	16.63	3.52
HAR-SPX-LR	26.74	19.17	4.17
HAR-VIX	24.81	15.78	3.33
HAR-VIX-LR	27.01	18.66	4.09
Panel HAR	22.39	15.02	3.08
Panel HAR-LR	21.82	14.10	2.86
Log-HAR	19.75	18.96	3.81
MIDAS	22.69	15.72	3.29
Panel MIDAS	23.11	14.48	3.01
Panel B: Portfolio based on averaged forecasts			
$\eta = 0$	26.25	13.67	2.78
Panel C: Portfolios based on combined forecasts $\delta = 12$			
$\eta = 1/2$	24.31	13.84	2.82
$\eta = 1$	23.13	14.22	2.90
$\eta = \infty$	23.37	21.02	4.26
Panel D: Portfolios based on combined forecasts $\delta = 48$			
$\eta = 1/2$	24.56	13.72	2.80
$\eta = 1$	23.35	13.95	2.85
$\eta = \infty$	23.47	21.02	4.26

Notes: We report summary measures of the model-based portfolios. 'WAD' refers to the monthly sum of absolute distances between the model-based portfolio weights and the weights of the infeasible post hoc portfolio reported in percentages. We report the time-series average of WADs. For the definition of turnover (TO), see Eq. (7). The WAD and TO are reported in percentages. The TC is reported in basis points. The evaluation period is from 2005:M1–2021:M12.

5.3. Portfolio performance relative to the benchmark

We measure the performance of the portfolios by their out-of-sample returns, volatilities, and Sharpe ratios. The Sharpe ratios are the average out-of-sample excess return divided by the out-of-sample volatility. The tables report annualized values by simple 12 and $\sqrt{12}$ scaling.

With the portfolio turnover and the effective trading cost estimates, we compute the portfolio returns in excess of the one-month Treasury bill rate, $R_{rf,m}$, as

$$R_{p,m}^j = \frac{W_m^j}{W_{m-1}^j} - 1 - R_{rf,m}. \tag{8}$$

Here, W_m^j is the wealth of the portfolio, which can be obtained as

$$W_m^j = W_{m-1}^j \cdot (1 + (w_m^j)' R_m) \cdot (1 - TC_m^j). \tag{9}$$

Table 6 shows the annualized returns of each low-volatility portfolio with and without trading costs. As before, the 12m-RV^d portfolio is the benchmark portfolio. We report the annualized return and test whether there is a significant difference between the return of the respective forecast-based portfolio and the benchmark. Inference is based on Newey–West standard errors. We also test for differences in Sharpe ratios and volatilities between the respective portfolios and the benchmark using tests proposed by Ledoit and Wolf (2008, 2011).

Without transaction costs, almost all HAR specifications, the MIDAS models, and the forecast combination-based portfolios obtain annualized Sharpe ratios significantly higher than the Sharpe ratio of the 12m-RV^d

Table 6
Returns of low-volatility portfolios.

	Without TC				With TC			
	Ret	Std	SR	Ret-Gain	Ret	Std	SR	Ret-Gain
Post hoc	12.74	12.70	1.00	–	12.14	12.71	0.96	–
Panel A: Portfolios based on individual forecasts								
12m-RV^d	10.52	13.86	0.76	0.00	10.28	13.87	0.74	0.00
4y-RV^d	10.68	14.06**	0.76	0.01	10.48	14.07**	0.74	0.06
6m-RV^d	10.72	13.77	0.78	0.27	10.41	13.79	0.76	0.20
1m-RV^d	10.86	13.50**	0.80	0.63**	9.87	**13.51****	0.73	−0.15
RM monthly, 12 months	10.55	13.87	0.76	0.03	10.31	13.88	0.74	0.03
RM monthly, 6 months	10.72	13.77	0.78	0.27	10.41	13.79	0.75	0.19
RM daily, 12 months	10.69	13.75	0.78	0.25	10.22	13.77	0.74	0.02
RM daily, 6 months	10.69	13.71	0.78	0.30	10.21	13.72	0.74	0.05
GJR-GARCH	10.80	13.69**	0.79	0.41	10.30	13.70**	0.75	0.15
Panel GJR-GARCH	10.62	13.77	0.77	0.17	10.20	13.79	0.74	−0.01
Factor GARCH	10.84*	13.93	0.78	0.27	10.46	13.94	0.75	0.14
Factor GARCH-VIX	10.83*	14.01*	0.77	0.20	10.44	14.02*	0.74	0.05
Factor GARCH-ΔHous	10.89*	14.04**	0.78	0.23	10.50	14.05**	0.75	0.08
Factor GARCH-TS	10.88*	14.06**	0.77	0.20	10.49	14.07**	0.75	0.06
MEM	**11.04**	13.78	0.80*	0.59*	10.56	13.80	0.77	0.34
Panel MEM	10.93	13.70**	0.80*	0.54*	10.48	13.71**	0.76	0.33
R-HAR	10.85	**13.45*****	0.81**	0.67**	10.29	13.46***	0.76	0.33
HAR	10.82	13.62**	0.79**	0.50**	10.40	13.64**	0.76	0.30
HAR-LR	10.96*	13.64**	0.80**	0.62**	10.47	13.66*	0.77	0.35
HAR-SPX	10.73	13.67*	0.79	0.36	10.31	13.69*	0.75	0.17
HAR-SPX-LR	10.89	13.55**	0.80**	0.62***	10.39	13.57**	0.77	0.34
HAR-VIX	10.86*	13.64**	0.80**	0.51**	10.45	13.66**	0.77*	0.34
HAR-VIX-LR	10.98*	13.56**	0.81***	0.71***	10.49	13.58**	0.77*	0.44*
Panel HAR	10.92	13.70**	0.80**	0.53**	10.55	13.72**	0.77	0.40
Panel HAR-LR	11.03**	13.60**	**0.81*****	**0.73*****	**10.69***	13.61**	**0.79****	**0.62****
Log-HAR	10.86	13.51***	0.80**	0.62**	10.40	13.53***	0.77	0.39
MIDAS	11.01**	13.71**	0.80**	0.62**	10.61	13.72**	0.77*	0.45*
Panel MIDAS	10.95*	13.67**	0.80**	0.58**	10.59	13.69**	0.77*	0.45*
Panel B: Portfolio based on averaged forecasts								
$\eta = 0$	10.80	13.71**	0.79**	0.40**	10.47	13.72**	0.76*	0.30*
Panel C: Portfolios based on combined forecasts $\delta = 12$								
$\eta = 1/2$	10.81	13.69***	0.79**	0.42**	10.47	13.70***	0.76	0.32*
$\eta = 1$	10.87*	13.69**	0.79**	0.49**	10.52	13.70**	0.77*	0.38*
$\eta = \infty$	10.86	13.46***	0.81**	0.67***	10.34	13.47***	0.77	0.37*
Panel D: Portfolios based on combined forecasts $\delta = 48$								
$\eta = 1/2$	10.81	13.69**	0.79**	0.43**	10.48	13.71**	0.76*	0.33*
$\eta = 1$	10.87*	13.69***	0.79**	0.49**	**10.53**	13.70***	0.77*	0.39*
$\eta = \infty$	**10.95***	**13.40*****	**0.82*****	**0.81*****	10.44	**13.42*****	**0.78***	**0.51****

Notes: Average annualized excess return (Ret), annualized standard deviation (Std), and Sharpe Ratio (SR). We perform two-sided tests of equal returns using Newey–West standard errors with three lags against the benchmark model 12m-RV^d. Sharpe ratio test according to Ledoit and Wolf (2008) and the volatility test according to Ledoit and Wolf (2011). In Panel A, numbers in bold represent the highest return, the lowest standard deviation, the highest SR, and the highest Ret-Gain. We jointly apply the same highlighting to all forecast combinations in Panels B, C, and D. We report all figures up to two decimals. The evaluation period is from 2005:M1–2021:M12.

* Statistical significance at the 10% level is indicated.

** Statistical significance at the 5% level is indicated.

*** Statistical significance at the 1% level is indicated.

benchmark. The increase in the Sharpe ratio is due to a significantly lower standard deviation and, in some cases, a significantly higher return. Among the forecast-based portfolios with the highest Sharpe ratios are the R-HAR, HAR-VIX, Panel HAR, Panel HAR-LR, and Log-HAR—i.e. the models that performed exceptionally well in out-of-sample forecasting (see Table 4). For those forecast-based portfolios, the WAD in Table 5 was among the lowest. That is, better forecast performance translates into low-volatility portfolios closer to the post hoc portfolio and hence a significant improvement relative to the benchmark. The portfolio based on the $\delta = 48$ and $\eta = \infty$ forecast combination, i.e., the BF-based portfolio, performs best among all portfolios. Recall that this combination places all weight on the best forecasting model each month. When accounting for transaction costs, the number of forecast-based portfolios that still achieve a significantly higher Sharpe ratio drops considerably. Nevertheless, the Sharpe ratios of the Panel HAR-LR and the BF-based portfolio are still significantly higher at the 5% level. Specifically, the Panel HAR-LR benefits from low turnover and low transaction costs.

Table 7
Tracking error and fees.

	Without TC			With TC		
	TE	Δ_1	Δ_{10}	TE	Δ_1	Δ_{10}
Panel A: Portfolios based on individual forecasts						
12m-RV^d	0.51	2.38	3.96	0.51	2.02	3.60
4y-RV^d	0.61	2.24	4.10	0.61	1.85	3.70
6m-RV^d	0.46	2.16	3.62	0.46	1.88	3.33
1m-RV^d	0.40	1.98	3.06	0.40	2.38	3.45
RM monthly, 12 months	0.50	2.34	3.93	0.51	1.99	3.57
RM monthly, 6 months	0.46	2.17	3.62	0.46	1.88	3.34
RM daily, 12 months	0.44	2.19	3.61	0.44	2.07	3.50
RM daily, 6 months	0.42	2.18	3.54	0.43	2.07	3.43
GJR-GARCH	0.44	2.08	3.41	0.44	1.98	3.32
Panel GJR-GARCH	0.49	2.26	3.71	0.49	2.08	3.54
Factor GARCH	0.52	2.07	3.74	0.52	1.85	3.51
Factor GARCH-VIX	0.54	2.08	3.88	0.54	1.88	3.67
Factor GARCH-ΔHous	0.55	2.03	3.87	0.55	1.83	3.66
Factor GARCH-TS	0.56	2.04	3.91	0.56	1.84	3.70
MEM	0.48	1.84	3.30	0.49	1.73	3.20
Panel MEM	0.43	1.94	3.29	0.43	1.79	3.14
R-HAR	**0.35**	1.99	**2.98**	**0.35**	1.95	2.95
HAR	0.42	2.04	3.27	0.43	1.87	3.11
HAR-LR	0.44	1.91	3.17	0.45	1.80	3.08
HAR-SPX	0.43	2.14	3.44	0.44	1.97	3.28
HAR-SPX-LR	0.45	1.97	3.11	0.46	1.87	3.03
HAR-VIX	0.43	2.01	3.28	0.44	1.82	3.08
HAR-VIX-LR	0.48	1.87	3.03	0.48	1.77	2.93
Panel HAR	0.42	1.95	3.30	0.42	1.73	3.08
Panel HAR-LR	0.39	**1.83**	3.03	0.39	**1.58**	**2.78**
Log-HAR	0.37	1.99	3.08	0.37	1.85	2.94
MIDAS	0.41	1.86	3.22	0.42	1.66	3.03
Panel MIDAS	0.41	1.92	3.23	0.42	1.69	3.01
Panel B: Portfolio based on averaged forecasts						
$\eta = 0$	0.43	2.07	3.44	0.43	1.81	3.18
Panel C: Portfolios based on combined forecasts $\delta = 12$						
$\eta = 1/2$	0.41	2.07	3.40	0.41	1.81	3.14
$\eta = 1$	0.40	2.00	3.33	0.41	**1.75**	3.08
$\eta = \infty$	0.36	1.98	2.99	0.36	1.90	2.92
Panel D: Portfolios based on combined forecasts $\delta = 48$						
$\eta = 1/2$	0.42	2.06	3.40	0.42	1.80	3.14
$\eta = 1$	0.41	2.00	3.33	0.41	**1.75**	3.08
$\eta = \infty$	**0.34**	**1.88**	**2.82**	**0.35**	1.80	**2.74**

Notes: The tracking error (TE) is calculated as the standard deviation of the difference between the post hoc portfolio return and the model-based portfolio return. Δ_γ is the annualized fee in percent an investor would be willing to pay for switching to the infeasible post hoc portfolio; see Eq. (11). In Panel A, numbers in bold represent the lowest tracking error and fees Δ_γ across all models. We jointly apply the same highlighting to all forecast combinations in Panels B, C, and D. We report all figures up to two decimals. The evaluation period is from 2005:M1–2021:M12.

To contextualize the reported gains in financial performance, we compute the gain in average yearly returns that an investor can achieve by switching from the benchmark portfolio to a forecast-based portfolio while taking the same level of risk as before. We define the return gain, $Ret - Gain$, as

$$Ret - Gain^j = \frac{Ret^j}{Std^j} Std^{(12m-RV^d)} - Ret^{(12m-RV^d)}. \quad (10)$$

Without accounting for transaction costs, Table 6 reveals that an investor can increase average annual returns by switching to the Panel HAR-LR-based portfolio or the BF-based portfolio by 0.73 or 0.81 percentage points. Even after trading costs, the gain of switching to those forecast-based portfolios remains economically sizable, with 0.62 and 0.51 percentage points.

5.4. Comparison with post hoc portfolio

Last, we compare the forecast-based portfolios with the infeasible post hoc portfolio. As Table 6 shows, there is still a considerable gap between the forecast-based and post hoc portfolios' performance. Because future volatility is not perfectly predictable, this gap is unavoidable.

To explain the gap, we first report the tracking error of each forecast-based portfolio with respect to the post hoc portfolio in Table 7. We define the tracking error as the (time-series) standard deviation of the difference between the return of the post hoc and the respective forecast-based portfolio. Confirming our previous results, the 'TE' column of Table 7 shows that the tracking errors of the Panel HAR-LR and the BF-based portfolios are (among) the

lowest and far below the tracking error of the benchmark portfolio.

Next, we use a utility framework to quantify the gap between the forecast-based portfolios and the post hoc portfolio. We follow Fleming et al. (2001, 2003) and compute the fee required to make a hypothetical investor indifferent with respect to the forecast-based portfolios and the post hoc allocation. Using a quadratic utility function with risk-aversion parameter γ, the monthly utility generated by a portfolio based on model j is given by

$$U_\gamma(R^j_{p,m}) = (1 + R^j_{p,m}/100) - \frac{\gamma}{2(1+\gamma)}\left(1 + R^j_{p,m}/100\right)^2.$$

This utility is compared to the utility obtained from the post hoc portfolio under a fee. Denoting the return of the post hoc portfolio by $R^{ph}_{p,m}$, we compute the maximum fee Δ^j_γ that an investor would be willing to pay in order to switch from portfolio j to the post hoc portfolio by solving

$$\sum_{m=1}^{M} U_\gamma(R^j_{p,m}) = \sum_{m=1}^{M} U_\gamma(R^{ph}_{p,m} - \Delta^j_\gamma). \tag{11}$$

Thus, Δ^j_γ is the fee required to make an investor indifferent between the post hoc and feasible allocations. The fee Δ^j_γ is low if the utility of portfolio j is close to the utility obtained under the infeasible post hoc portfolio. Table 7 reports the fee Δ^j_γ in annualized percentage points for $\gamma = 1$ and $\gamma = 10$, which aligns with the related literature (Fleming et al., 2001).

Generally, we see that the required fee Δ_γ decreases when trading costs are introduced. This implies that while the post hoc portfolio obtains the highest net performance, it is comparably expensive to maintain. The more risk-averse investors ($\gamma = 10$) demand higher fees across all models.

Independently of transaction costs, for investors with low risk aversion ($\gamma = 1$), the fee is the lowest for the Panel HAR-LR-based portfolio. Specifically, with transaction costs and for $\gamma = 1$, the fee of the Panel HAR-LR-based portfolio is 1.58 percentage points, which is considerably lower than the fee of 2.02 for the benchmark portfolio. Thus, investors with low risk aversion are happy to base their investment decision on a single model. In contrast, for investors with high risk aversion ($\gamma = 10$), the fee associated with the BF-based portfolio is the lowest. This suggests that investors with high risk-aversion prefer the low tracking error in combination with the diversification that comes with the BF-based portfolio, which selects the best forecasting model for each stock.

6. Conclusion

We investigated the benefits of using volatility models for implementing low-volatility portfolios for the US stock market. Low-volatility portfolios are typically constructed by sorting stocks based on historical volatility. In contrast, we employed a variety of time-series models based on intraday data, including RiskMetrics, GARCH, HAR, and MIDAS regressions. We compared their forecasting and portfolio performance against a benchmark portfolio based on a commonly used backward-looking measure of volatility.

We focused on one-month-ahead predictions and evaluated forecast performance in a large cross-section of 1699 stocks. Our empirical analysis revealed compelling results, illustrating the superior out-of-sample forecast performance of HAR-type models and forecast combinations. We found that forecast-based low-volatility portfolios significantly enhanced the financial performance relative to the benchmark portfolio. Our findings demonstrate that a panel HAR-based portfolio and a portfolio based on the best stock-specific forecasting model have the best financial performance. These results are statistically significant and robust to accounting for estimated effective trading costs.

Even though our set of time-series models was large, future research might still look at alternative models to improve low-volatility investments. Further, in this study, we did not include enhanced volatility-timing strategies. For example, Kirby and Ostdiek (2012) propose volatility-timing strategies with additional parameters that control timing aggressiveness. Last, the inverse-volatility allocations that we considered in this paper do not take into account covariances between returns. Considering non-zero covariances introduces possible diversification opportunities to the portfolio allocation. However, the number of covariances grows quadratically in the cross-section of assets, which provides an immediate high-dimensional estimation challenge. This statistical problem is an active field of research with many contributors—including Ledoit and Wolf (2017), Goto and Xu (2015), and Callot et al. (2021).

Data and code availability

The main data used in this study is proprietary and comes from the Trade and Quote (TAQ) database, the Center for Research in Security Prices (CRSP) via Wharton Research Data Services (WRDS) and a Bloomberg terminal. The data must be purchased separately through WRDS, and redistribution is not permitted. The source code used for this study is publicly available at: https://github.com/onnokleen/ReplicationLowVolatilityInvesting.

CRediT authorship contribution statement

Christian Conrad: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Onno Kleen:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rasmus Lönn:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.08.006.

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